Problem

Title

Perfect Numbers up to N

Topics

Divisors, Number Theory, Nested Loops, Conditional Statements, Mathematical Classification

Difficulty Level

Easy

Problem Detailed Statement (Scenario Based)

During a mathematical foundations session, students are introduced to the concept of **Perfect Numbers**.

A **Perfect Number** is a positive integer that is equal to the sum of its proper divisors (excluding the number itself).

A proper divisor of a number is any positive divisor other than the number itself.

For example:

- 6
  Proper divisors: 1, 2, 3
  Sum = 1 + 2 + 3 = 6 ✓
- 28
  Proper divisors: 1, 2, 4, 7, 14
  Sum = 28 ✓

However,

- 10
  Proper divisors: 1, 2, 5
  Sum = 8 ✗ (Not Perfect)

Now, the trainer provides a positive integer **N**. Students must:

1. Iterate through all numbers from 1 to N.
2. For each number:

    - Find all proper divisors using loop logic.
    - Calculate their sum.
    - Check whether the sum equals the original number.
3. Print all Perfect Numbers up to N.

Students must:

- Use nested loops (outer loop for numbers, inner loop for divisor checking).
- Avoid using built-in divisor libraries.
- Handle edge cases properly (N < 2).
- Maintain optimized divisor checking (optional: iterate till n/2).

If no Perfect Number exists up to N, print `-1`.

This problem enhances:

- Divisor logic
- Nested loop implementation
- Mathematical reasoning
- Classification of numbers

Input Format

A single integer:

```
N
```

Constraints:

```
1 ≤ N ≤ 10^5
```

Output Format

- Print all Perfect Numbers up to N in ascending order separated by space.
- If none exist, print:

```
-1
```

Examples

Example 1

Input:

```
30
```

Output:

```
6 28
```

Example 2

Input:

```
5
```

Output:

```
-1
```

Example 3

Input:

```
500
```

Output:

```
6 28 496
```